APRM, EPP-I, 2025-26, GLA U Mathura
Module-1 (Verbal Skills)
Reading Comprehension-II
Passage 1: Minds, Machines, and Models of Thought

When Alan Turing proposed his now-famous question — "Can machines think?" — he did not merely initiate a technical inquiry but a psychological one. His 1950 essay, Computing Machinery and Intelligence, reframed intelligence as a matter of behavioral equivalence rather than biological essence. To think, in Turing's pragmatic sense, was to act indistinguishably from a human thinker. In the decades since, this notion has shaped both computer science and psychology, transforming how we model the mind and measure intelligence.

In early cognitive science, the mind was imagined as an information processor — a system that encoded, stored, and retrieved data according to computational rules. Psychologists borrowed metaphors from computer science: the brain as hardware, the mind as software, and thought as computation. This "computational theory of mind" became a unifying framework for explaining perception, memory, and reasoning. Computers offered not only metaphors but also methods: by programming models of mental processes, scientists could test theories of cognition with an unprecedented level of precision.

However, as artificial intelligence (AI) advanced, the simplicity of this analogy began to fray.
The early "symbolic" AI systems of the 1960s and 1970s — built on formal logic and rule-based reasoning — captured some aspects of human problem-solving but failed to replicate the flexibility and intuition of real human cognition. Psychologists observed that people rarely reason by applying explicit rules; they rely on heuristics, emotions, and context. Computers, for all their logical rigor, often stumbled where humans excelled — in tasks demanding common sense, empathy, or improvisation.

The rise of connectionism in the 1980s marked a shift toward more biologically inspired models. Neural networks — distributed systems that learn through patterns of activation — offered a way to mimic the adaptive, nonlinear nature of the brain. In these models, intelligence emerges not from predefined rules but from learning statistical regularities across vast data sets. The approach mirrored discoveries in cognitive psychology: that much of human thought is associative, not algorithmic. Still, this resemblance was partial. Artificial neural networks learn correlations; human minds also learn meanings. The gap between statistical learning and conscious understanding remains one of the central puzzles of both AI and psychology.

Contemporary research at the interface of these fields explores precisely that gap. Cognitive architectures such as ACT-R and SOAR attempt to integrate symbolic and connectionist elements, modeling how perception, memory, and decision-making interact dynamically.
Meanwhile, neuroscientists use computational tools to simulate how patterns of neural firing might correspond to mental representations — bridging biological and computational levels of explanation. The resulting dialogue is two-way: psychology informs computer science by providing models of cognition, and computer science returns the favor by offering testable, quantitative frameworks for mental phenomena.

Yet this convergence also raises philosophical and ethical questions. If cognition can be modeled computationally, does that imply that minds are reducible to algorithms? Many psychologists resist this reduction, arguing that subjective experience — the "what it feels like" to think — escapes formalization. A program might simulate the behavior of understanding without possessing the experience of it. This distinction echoes the famous "Chinese Room" argument proposed by philosopher John Searle: that syntax alone (symbol manipulation) does not yield semantics (meaning). For computer scientists, however, this critique invites rather than closes inquiry — suggesting that future models may need to integrate embodiment, affect, and environment into computational accounts of mind.

An even deeper implication concerns our own self-conception. If human intelligence can be replicated in silico, what does that say about the uniqueness of human psychology? Some argue that building artificial minds helps us understand natural ones: simulating intelligence exposes the mechanisms of cognition by reconstruction.

Others fear that treating minds as programmable systems risks trivializing personhood — reducing thought to computation and emotion to code. Between these poles lies a middle ground: the recognition that computational models are not mirrors of the mind but instruments for probing its structure. They are ways of thinking about thinking.

As machine learning systems now emulate creative, linguistic, and emotional behavior, the boundary between psychological theory and computational practice grows increasingly porous. The psychologist's experiment and the computer scientist's model often target the same question: how does intelligence — human or artificial — arise from matter and interaction? The collaboration between these fields has already yielded insights into attention, perception, and decision-making that neither could achieve alone. Whether or not machines ever truly "think," their design continues to teach us what it means when we do.

Comprehension Questions:

1. The author implies that early symbolic AI systems failed to match human cognition primarily because:
A. They lacked sufficient computing power.
B. They relied on rigid logical rules unlike the flexible, context-sensitive processes of human thought.
C. They were based on inaccurate neural models.
D. Psychologists rejected computational metaphors.

2. Which of the following cognitive architectures is mentioned in the passage?
A. AlphaZero
B. LIDA
C. ACT-R
D. Deep Blue

3. The tone of the passage can best be described as:
A. Analytical and exploratory, balancing enthusiasm with philosophical caution.
B. Skeptical and dismissive of computational models.
C. Optimistic and celebratory about machine superiority.
D. Detached and purely historical.

4. Searle's "Chinese Room" argument is invoked to illustrate that:
A. Human cognition cannot be explained scientifically.
B. Symbol manipulation alone does not equate to genuine understanding.
C. Computers will inevitably become conscious.
D. Semantics can be reduced to syntax with enough data.

5. What does the author suggest about the relationship between computer science and psychology today?
A. They operate in isolation, addressing unrelated questions.
B. They form a reciprocal relationship, each providing methods and insights to the other.
C. Psychology now depends entirely on computer models.
D. Computer science has replaced psychology as the main science of the mind.

6. The discussion of embodiment and affect implies that:
A. Future computational models may need to incorporate emotional and physical context to approach true understanding.
B. Emotions are obstacles to rational computation.
C. Human cognition can be entirely separated from the body.
D. Computational models already replicate human feeling accurately.

7. Which statement best summarizes the central argument of the passage?
A. The interplay between computer science and psychology deepens our understanding of both artificial and human intelligence.
B. Psychology should reject computational methods in favor of phenomenology.
C. Artificial intelligence will soon render human cognition obsolete.
D. The computational theory of mind has been conclusively disproved.

Passage 2: The Fragile Architecture of Scientific Truth

The notion that science provides us with objective truth has long been regarded as one of modernity's most comforting certainties. The scientist, armed with empirical data and rational methodology, is thought to reveal the world as it truly is. Yet, this image of science as a transparent window to reality has been steadily eroded by a series of philosophical critiques that began in the twentieth century and continue to reverberate today. The tension between the **ideal of objectivity** and the **social embeddedness of science** is not merely an academic concern—it defines the very credibility of scientific knowledge in a world saturated with competing claims to truth.

In the early part of the twentieth century, logical positivists such as Carnap and Schlick sought to secure science upon an unshakable foundation. They proposed that scientific statements derive their meaning from direct empirical verification. Observation was king, and the scientist was to be the neutral observer—stripped of bias, emotion, and cultural prejudice. However, this vision soon proved untenable. As Quine famously argued, no observation is ever theory-free. What one perceives is always filtered through a conceptual framework, through linguistic habits, through a set of presuppositions that shape what counts as evidence in the first place. Thus, the very act of observation is not a passive reception of facts but an interpretive act.

Thomas Kuhn deepened this insight in his *Structure of Scientific Revolutions* (1962), where he introduced the now-famous notion of a "paradigm." For Kuhn, science progresses not linearly, but episodically—through revolutions in which one dominant framework is replaced by another that is often **incommensurable** with it. The transition from Newtonian mechanics to Einsteinian relativity, for example, was not simply the addition of new data to an existing framework but a radical transformation of the conceptual scaffolding that made scientific understanding possible. Kuhn's critics accused him of relativism, of suggesting that truth in science is merely a matter of consensus. Yet Kuhn himself denied this charge, insisting instead that while paradigms are historically contingent, they are not arbitrary. Scientific revolutions occur because new paradigms solve problems that the old ones cannot.

If Kuhn destabilized the notion of objective progress, Paul Feyerabend went further, arguing in *Against Method* (1975) that there is no single scientific method at all. He provocatively declared that "anything goes" in the advancement of science. Feyerabend's intention, however, was not to promote chaos, but to highlight the **creative, pluralistic, and historically situated** character of scientific inquiry. What counts as rational in one era may appear irrational in another. Galileo's reliance on rhetoric and persuasion, for instance, was condemned

by his contemporaries as unscientific, yet it was instrumental in ushering in the modern scientific worldview. Feyerabend's anarchism was thus less a rejection of reason than a plea to recognize the diversity of ways in which reason operates.

More recent thinkers, such as Bruno Latour and Donna Haraway, have turned the lens of sociology and feminism on science, challenging the assumption that knowledge production occurs in a vacuum. Latour's *Actor-Network Theory* suggests that scientific facts are not simply "discovered" but are **constructed** through complex networks involving human and non-human actors—laboratories, instruments, funding bodies, journals, and political institutions. Haraway, in her seminal essay "Situated Knowledges," argues that all knowledge is "partial, locatable, and critical." The supposed "view from nowhere" that traditional science claims to occupy is, for her, an illusion that obscures the situated nature of all seeing. Objectivity, she contends, is not the elimination of perspective, but the conscious acknowledgment of it.

What emerges from these debates is not a dismissal of science but a more nuanced understanding of it. To acknowledge the social, historical, and interpretive dimensions of science is not to render it meaningless. On the contrary, it is to situate it within the web of human activity, where its achievements can be appreciated not as revelations of eternal truth but as evolving practices of understanding. Science remains our most powerful means of engaging with the material world, but its authority must rest on transparency and reflexivity, not on the myth of pure objectivity. In the end, the philosophy of science invites us to see scientific truth not as a fortress but as a **fragile architecture**—resilient yet revisable, built through collaboration, argument, and imagination. The fragility of this architecture does not weaken science; rather, it humanizes it. It reminds us that knowledge, like the beings who pursue it, is always in motion—forever provisional, forever incomplete, and yet endlessly striving toward greater coherence.

Comprehension Questions:

1. The author's central argument about science can best be summarized as:
A. Science has failed to achieve objectivity and should be replaced by more subjective methods of inquiry.
B. Science is a human practice that, while striving for truth, must recognize its historical and social contingency.
C. Modern science remains the only reliable path to absolute truth.
D. The primary problem with science is its reliance on empirical observation.

2. According to the passage, Quine's critique of logical positivism emphasizes that:
A. Observation is independent of theory and therefore reliable.
B. Meaning in science comes solely from empirical verification.
C. Observation is always shaped by theoretical and linguistic frameworks.
D. Scientific statements must be grounded in pure logic.

3. The tone of the passage can best be described as:
A. Reverent and celebratory of scientific triumphs.
B. Cynical toward all forms of scientific reasoning.
C. Reflective and critical, yet respectful of science's importance.
D. Detached and purely descriptive without evaluative commentary.

4. What does the author imply about Kuhn's concept of paradigms?
A. Paradigm shifts are random events unrelated to evidence.

B. Paradigm changes are purely social phenomena without rational justification.
C. Paradigms evolve when they better address unsolved problems within science.
D. Paradigms represent political rather than epistemological transformations.

5. Feyerabend's notion of "anything goes" is presented in the passage primarily to:
A. Advocate for the abandonment of rationality in scientific inquiry.
B. Emphasize the creative and pluralistic nature of scientific progress.
C. Argue that science should adhere strictly to established procedures.
D. Illustrate the superiority of pre-modern science over modern science.

6. Haraway's "situated knowledges" challenges traditional objectivity by asserting that:
A. All knowledge must eliminate perspective to be credible.
B. Knowledge is inevitably partial and shaped by the position of the knower.
C. Feminist perspectives distort the neutrality of science.
D. Objectivity can only be achieved through mathematical precision.

7. Which of the following best captures the metaphor of "fragile architecture" in the final paragraph?
A. Scientific knowledge is structurally unstable and therefore unreliable.
B. The achievements of science rest on delicate yet adaptable human efforts.
C. Science must protect its foundations from all external criticism.
D. The pursuit of knowledge inevitably leads to collapse.

Passage 3: The Boundaries of Explanation

The ambition of science has always been to explain. From the earliest natural philosophers who sought to uncover the principles behind celestial motion, to today's neuroscientists probing the workings of consciousness, explanation has been the common currency of intellectual progress. Yet, if explanation is science's highest goal, it is also its most elusive. What it means to **explain** something — to make sense of it — turns out to be as much a philosophical as an empirical question.

In the nineteenth century, many believed that explanation consisted in *reduction*: to explain a phenomenon was to show how it could be derived from more fundamental laws. This view found its purest expression in classical physics, where the motions of planets and pendulums alike were accounted for by Newton's laws. If the behavior of matter could be explained by simple principles, then, it seemed, all natural phenomena could ultimately be reduced to mechanics. Chemistry, biology, and even human psychology were thought to be, in principle, extensions of physical law. But, as the sciences matured, this ideal of reductionism began to fracture. In the mid-twentieth century, philosophers such as Ernest Nagel tried to preserve the dream by formulating **bridge laws** — statements that would link the vocabulary of one science (say, biology) to that of another (physics or chemistry). Yet, these bridges proved fragile. Biological systems, for instance, exhibit forms of organization, adaptation, and emergence that resist full translation into the language of physical particles and forces. The more one descends into the microscopic, the more one risks losing sight of the phenomena that make life *life-like*.

Enter the notion of **emergence** — the idea that higher-level patterns and behaviors can arise from lower-level components but are not reducible to them. The flight of a flock, the self- organization of cells, the dynamics of

ecosystems: each of these can be modeled in terms of underlying particles and forces, yet their coherence depends on relationships and feedback loops that operate at higher levels of complexity. To explain them, one must invoke concepts like "function," "purpose," or "information," which belong not to physics, but to biology, systems theory, and cybernetics.

The recognition of emergence did not end reductionism but rather transformed the debate. The question ceased to be whether higher-level explanations could, *in principle*, be reduced to lower- level ones, and became whether reduction is the **best way** to understand the world. For some philosophers, such as Daniel Dennett, multiple "levels of explanation" coexist legitimately. An event like the decision to raise one's hand can be described neurologically, psychologically, or socially — and each level captures truths invisible to the others.

However, the pluralism of explanation also introduces tension. If different explanations are all valid, how do we judge which is better or more fundamental? The philosopher Nancy Cartwright famously argued that scientific laws "lie" — not because they are false, but because they simplify the messy contingencies of the real world. Laws, she suggests, work locally, within specific domains; outside those domains, they lose their grip. In this sense, the universe is less a single, unified machine than a **patchwork** of models, each effective within its proper boundaries.

This patchwork vision unsettles the classical dream of unity in science. If reality resists reduction to a single framework, then the goal of science may not be to find the ultimate explanation, but to weave together many partial ones — each illuminating different aspects of a complex, layered world. The biologist's concept of adaptation does not compete with the physicist's equations; it simply addresses a different question. Explanation, then, becomes a matter of perspective: what counts as a good explanation depends on what we wish to understand and why.

Yet, some critics worry that this turn toward pluralism risks undermining science's authority. If every explanation is merely "situated," does that not lead to relativism — to the idea that all accounts are equally valid? The challenge is to preserve both humility and rigor: to recognize the contextuality of knowledge without surrendering to arbitrariness. Perhaps, as philosopher Philip Kitcher proposes, objectivity lies not in escaping human perspective but in **achieving coherence** among many perspectives. Explanation, in this view, is not the uncovering of eternal laws but the ongoing work of integration — the art of seeing how different stories about the world can fit together without collapsing into one.

In this light, the boundaries of explanation are not barriers but frontiers — shifting, porous, and generative. The philosopher's task is not to close them but to navigate them: to remind us that every act of understanding, however precise, is also an invitation to look again, and more deeply, at what remains unexplained.

Comprehension Questions:

1. The author suggests that the traditional reductionist view of explanation is limited because:
A. It ignores the need for mathematical precision.
B. It oversimplifies complex phenomena by forcing them into lower-level terms.
C. It overemphasizes social and cultural influences in science.
D. It focuses too heavily on biological rather than physical systems.

2. According to the passage, emergence refers to phenomena that:
A. Can always be derived directly from physical laws.
B. Depend on higher-level relationships that cannot be fully captured by lower-level descriptions.
C. Violate the principles of physics and chemistry.
D. Replace the need for empirical observation.

3. The tone of the passage can best be described as:

A. Dogmatic and dismissive of scientific realism.
B. Analytical and balanced, emphasizing nuance and integration.
C. Polemical, aiming to discredit reductionist science.
D. Skeptical and cynical toward the scientific enterprise.

4. What does Nancy Cartwright's claim that "scientific laws lie" imply?
A. Science deliberately misrepresents reality for political gain.
B. Scientific theories are useless outside controlled experiments.
C. Scientific models simplify reality and are context-dependent rather than universally true.
D. The concept of natural law should be abandoned entirely.

5. The metaphor of science as a "patchwork of models" conveys that:
A. Science lacks coherence and cannot be trusted.
B. Different scientific theories can coexist, each valid within certain limits.
C. The ultimate goal of science is to eliminate competing explanations.
D. Only unified theories are genuinely scientific.

6. The passage implies that pluralism in explanation requires:
A. Abandoning all claims to objectivity.
B. Integrating multiple perspectives while maintaining coherence.
C. Rejecting the use of reductionist models altogether.
D. Treating all explanations as equally valid regardless of evidence.

7. Which of the following statements best expresses the passage's concluding insight?
A. Explanation is a static endpoint once sufficient knowledge is acquired.
B. The philosopher's task is to establish permanent boundaries between disciplines.
C. The search for explanation is an ongoing, open-ended process that invites deeper inquiry.
D. True explanation requires eliminating ambiguity and interpretation.